

Department of Economics
University of Oklahoma
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EDUCATION

- 2025 Ph.D. Candidate in Economics, University of Oklahoma, Norman, OK
Committee: Dr. [Joan Hamory](#) (Chair), Dr. [Jayash Paudel](#), Dr. [Tyler Ransom](#), and Dr. [Joseph Ripberger](#)
- 2017 B.A. in Economics and English Literature, Truman State University, Kirksville, MO
Honors: Magna Cum Laude and Departmental Honors in Economics
Member: Phi Beta Kappa

RESEARCH FIELDS

- Primary: Environment and Ecology Economics, Agricultural Economics
Secondary: Development Economics, Public Economics

JOB MARKET PAPER

"Agricultural trade policy and water quality: Evidence from the Mexican Hass Avocado Import Program" (Sole Author), Under Review, *American Journal of Agricultural Economics*

Abstract: This paper provides empirical evidence that the 2016 Mexican Hass Avocado Import Program amendment led to improved water quality due to Mexican avocado farmers incorporating cleaner production methods and applying less harmful pesticides to their avocados. Two-way fixed effects difference-in-differences analysis shows that the policy improved surface water quality, reducing biochemical oxygen demand (BOD) by approximately 12 percent and chemical oxygen demand (COD) by 11 percent. Further evidence suggests the amendment led to a 2 percent decrease in nitrate pollution and a nearly 1 percent increase in nitrites - likely stemming from cleaner avocado practices and increased avocado production.

WORKING PAPERS

"Fracking and Tracking: The Effects of Oil and Natural Gas Well Locations on the Housing Market" (with [Brent Norwood](#), [Samantha R. Johnson](#), and [Ahmed El Fatmoui](#)), Under Review, *Journal of Regional Science*

Abstract: Hydraulic fracturing or "fracking" has made it possible to produce vast new quantities of oil and natural gas causing states such as Colorado, Texas, and Oklahoma to dramatically increase their number of oil and natural gas wells. Using data from the U.S. Department of Homeland Security and Zillow's ZTRAX, we estimate the effect of hydraulically fractured

oil and natural gas well sites on both urban and rural residential home prices between 2000 and 2018. Utilizing ArcGIS, we create varying buffer zone sizes around well sites to explore how average home prices change before and after a well opens and apply a ZIP code-level fixed effects model, a household-level fixed effects model, a repeat sales model, and a spatial differences-in-differences (SDID) approach. Our results show that homes within 0.5 miles of a well have a 4.5% increase in selling price and homes between 0.5-1 mile of a well see a 2.5% increase in selling price compared to properties located more than 2 miles away from a well.

“Guaconomics: Avocado Price Shocks and Crime Rates in Mexico” (Sole Author)

Abstract Avocado trade between Mexico and the United States has exponentially burgeoned since 1997. Politicians have a growing concern about avocado production leading to more harm than good based on drug cartel involvement. Using data on 460 municipalities from 1997 to 2019, I exploit exogenous changes in U.S. weather deviations to show that increased Mexican avocado producer prices differentially decreased homicide in municipalities more climatically suited to grow avocados. This is because of increased agricultural employment from exponential growth in avocado production. However, there are different time heterogeneous effects. I find evidence that after the war on drugs took place in 2006, higher prices differentially increased homicide as well as other crime indicators such as property crime, cattle theft, and reported threats. The heterogeneous effects stem from increased drug cartel presence in the avocado industry.

“The Effects of Unbalanced FDI Inflow from Trade on the Gender Wage Gap in Mexico” (Sole Author)

WORKS IN PROGRESS

- 2024 “Fair for All Sizes? The Impacts of Wayfair on Small to Medium Sized Retail Firms’ Sales and Sales Tax ” (Sole Author)
- 2024 “Fracking and Water Contamination in Oklahoma” (With Brent Norwood and Ahmed Fatmoui)
- 2024 “Impact of Groundwater Quality on Infant Health: An Event Study in Mexico” (Sole Author)

PRESENTATIONS

- 2025 *Forthcoming* - Allied Social Science Association in San Francisco, CA (January)
- 2024 *Forthcoming* - Southern Economic Association in Washington D.C. (November)
- Forthcoming* - Missouri Valley Economic Association in Kansas City, MO (October)
- OU Ph.D. Economics Conference, Norman, OK (May)
- 2023 Southern Economic Association in New Orleans, LA (November)

FELLOWSHIPS AND AWARDS

2024	Alex Kondonassis Development Award Winner, University of Oklahoma
2023	Roberson Research and Travel Grant, University of Oklahoma
2022	Chong Liew Summer Research Award, University of Oklahoma
2021	Chong Liew Outstanding 1st Year Graduate Student Award, University of Oklahoma
2021	Bridwell Institute's Graduate Economic Freedom Index Colloquium Grant, Southern Methodist University

TEACHING EXPERIENCE

Instructor

2024	(Spring) Principles of Economics-Macro (ECON 1113-995), University of Oklahoma (Spring) Principles of Economics-Macro (ECON 1113-795), University of Oklahoma - For Concurrent High School Students
2023	(Fall) Principles of Economics-Micro (ECON 1113-995), University of Oklahoma (Fall) Principles of Economics-Micro (ECON 1113-795), University of Oklahoma - For Concurrent High School Students (Summer) Government Relations-Business (ECON 3713), University of Oklahoma
2022	(Fall) Elements of Statistics (ECON 2843), University of Oklahoma (Summer) Principles of Economics-Micro (ECON 1123), University of Oklahoma

Graduate Teaching Assistant

2020-Present	Econometric Analysis (ECON 4223), International Trade (ECON 3613), Labor Economics (ECON 3513), Health Economics (ECON 3523), Elements of Statistics (ECON 2843), Principles of Economics-Micro (ECON 1123)
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SKILLS

Coding:	R, Stata, Julia, ArcGIS Pro, Git, $\LaTeX$, SQL, Python, Office
Others:	Reduced form modeling, machine learning, web scrapping, discrete choice modeling
Languages:	English (Native), Spanish (Advanced)

REFERENCES

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